CHAPTER 2

Limits of sequences

S14, S16, S19, S20

Before you begin · Inequalities and geometric sequences

Without a calculator, in six minutes. Answer briefly. Use calculation with fractions and powers. Then choose the targeted follow-up indicated under the questions.

1. Bound $\frac{1}{n+1}$ for $n \geq 1$.
2. Compare 0.7^{n+1} and 0.7^n for $n \geq 0$.
3. Give the sign of $\sqrt{n^2+1} - n$.
4. Rationalise $\sqrt{x} - 1$.
5. Is an increasing sequence necessarily bounded above?
6. Give an example of a bounded non-constant sequence.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Take the reciprocals of two positive numbers; R16 recalls the effect of the sign. **2.** Write the next term as 0.7 times the previous; R02. **3.** Compare the positive squares before the roots. **4.** Multiply by the conjugate and expand its product. **5.** Try the sequence with general term n; distinguish variation and bound. **6.** Test the powers of -1 over six indices; R02. The R questions serve as review prompts with the teacher, even if their collective routine is planned for later.

Objectives

Distinguish convergence and divergence; calculate a limit; use comparison, the squeeze theorem and monotone convergence.

Course notes

Definitions

$u_n \to \ell \in \mathbb{R}$ means that any open interval containing ℓ contains all terms from a certain index. We say that (u_n) converges to ℓ. $u_n \to +\infty$ means that for every real number A, there is an index N such that $n \geq N \Rightarrow u_n \geq A$. $u_n \to -\infty$ means that for every real number A, there is N such that $n \geq N \Rightarrow u_n \leq A$.

A sequence may have no limit, not even an infinite one: $(-1)^n$ oscillates between -1 and 1. Every convergent sequence is bounded, but the converse is false.

Operations and comparisons

For two convergent sequences with limits a, b, the sum tends to $a+b$, the product tends to ab and the quotient to $\frac{a}{b}$ if $b \neq 0$. Rules involving infinity are used while checking signs. The forms $+\infty - \infty$, $0 \times \infty$, $\frac{\infty}{\infty}$, $\frac{0}{0}$ do not allow a direct conclusion.

Example. $\frac{3n^2-n+1}{2n^2+4} = \frac{3-\frac{1}{n}+\frac{1}{n^2}}{2+\frac{4}{n^2}} \to \frac{3}{2}$. For $\sqrt{n^2+n}-n$, multiplying by the conjugate gives $\frac{1}{\sqrt{1+\frac{1}{n}}+1} \to \frac{1}{2}$.

Theorems

If $u_n \leq v_n$ from a certain index onwards and $u_n \to +\infty$, then $v_n \to +\infty$. If $a_n \leq u_n \leq b_n$ from a certain index and $a_n, b_n \to \ell$, then $u_n \to \ell$ (squeeze theorem). Every increasing sequence bounded above converges. Every decreasing sequence that is bounded below converges. This theorem is assumed.

Proof by comparison. For every A, the terms of u exceed A from some index onwards N_1. The inequality $v_n \geq u_n$ holds beyond some index N_2. For $n \geq \max(N_1, N_2)$, $v_n \geq A$.

Proof: increasing and not bounded above. For every A, the absence of an upper bound gives N such that $u_N > A$. Since it is increasing, $n \geq N \Rightarrow u_n \geq u_N > A$, hence $u_n \to +\infty$. Not being bounded above is not enough on its own.

Geometric sequences

$$q^n \longrightarrow \begin{cases} 0 & \text{if } -1 < q < 1, \\ 1 & \text{if } q = 1, \\ +\infty & \text{if } q > 1. \end{cases}$$

If $q \leq -1$, (q^n) has no limit: alternation for $q = -1$; even and odd subsequences of opposite signs and with unbounded moduli for $q < -1$.

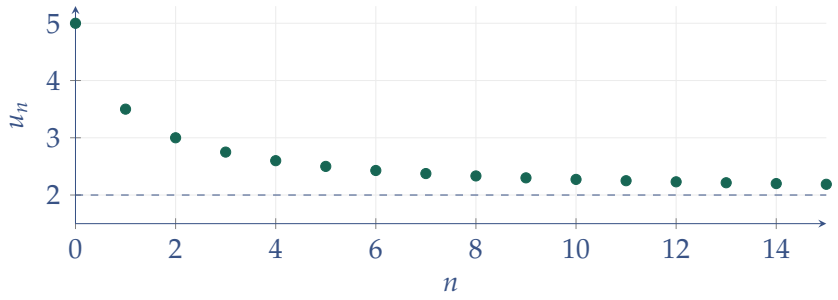
Bernoulli and geometric limits. Fix $a \geq 0$. Let us prove $P(n)$: "$(1+a)^n \geq 1+na$ ". At index 0, both sides equal 1. Fix $n \geq 0$ and suppose $P(n)$ true. Since $1+a > 0$,

$$(1+a)^{n+1} \geq (1+na)(1+a) = 1 + (n+1)a + na^2 \geq 1 + (n+1)a.$$

This is $P(n+1)$ since $na^2 \geq 0$. By induction, the inequality holds for every $n \geq 0$. If $q > 1$, take $a = q - 1 > 0$ and compare to $1 + na$. If $0 < q < 1$, apply this result to $\frac{1}{q}$, then take reciprocals. For $-1 < q < 0$, $|q^n| = |q|^n \to 0$ and squeezing concludes. The case $q = 0$ is handled from index 1.

Recursively defined sequence

For $u_{n+1} = f(u_n)$, establish a bound and monotonicity to prove convergence. If f is continuous at the limit ℓ, taking the limit gives $\ell = f(\ell)$ (continuity discussed in Chapter 4). Solving this equation never proves, on its own, that the sequence converges.



The points get closer to 2 without reaching it: $u_n = 2 + \frac{3}{n+1}$.

Give meaning to the limits

2.1 A band and all subsequent indices

First use: S14.

Definition · Converging to ℓ

We say (u_n) converges to ℓ if, for any precision $\varepsilon > 0$, there exists an index N such that $n \geq N$ implies $|u_n - \ell| < \varepsilon$. The index depends on the precision, but once chosen, it works for all the following terms.

Worked example · Calculating a valid index

For $u_n = 2 + \frac{3}{n+1}$, $|u_n - 2| = \frac{3}{n+1}$. The precision $0{,}1$ requires $n + 1 > 30$, so $n \geq 30$. Index 29 does not work: the gap there is exactly $0{,}1$. For any precision, every N such that $N + 1 > 3/\varepsilon$ works.

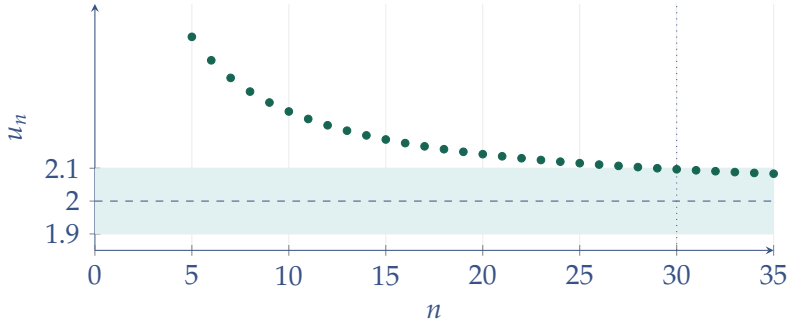


Figure 2.1. Band $]1{,}9;2{,}1[$: all terms from index $n \geq 30$ remain there.

A1 · Immediate application For $v_n = 5 - \frac{2}{n+1}$, determine the first index from which $4{,}95 < v_n < 5.05$. Check the previous index and explain why all later indices work.

A2 · Counterexample "There is a term in each band around 0, so the sequence converges to 0." Refute with a sequence whose even terms are zero and whose odd terms are 1.

2.2 Transform before concluding

First use: S16 and S19.

Worked example · A form does not determine a limit

The three quotients n/n, n/n^2 & n^2/n all have the form ∞/∞. Yet their limits are 1, 0 and $+\infty$. For $\frac{3n^2-n+1}{2n^2+4}$, dividing numerator and denominator by n^2 gives the limit $\frac{3}{2}$. The transformation is valid for $n \geq 1$, which is enough for a limit.

A3 · Choose before calculating Determine the limits of $\frac{2n+3}{n^2+1}$, $\sqrt{n^2+n}-n$ and $\frac{(-1)^n}{n+1}$. Name, for each, the decisive transformation or theorem.

A first step: In the second expression, multiply by the conjugate. In the third, bound the absolute value.

2.3 Converge, then identify

First use: S20; identification by continuity after S32.

Property · Monotone convergence · accepted without proof

An increasing sequence that is bounded above converges to a real number. The theorem says neither that it reaches its limit, nor that its limit is any one of its upper bounds.

Worked example · A complete chain

Let $u_0 = 1$ and $u_{n+1} = \sqrt{2+u_n}$. The interval $[1;2]$ is stable because $1 \leq u_n \leq 2$ implies $\sqrt{3} \leq u_{n+1} \leq 2$. On this interval, $2+x-x^2 = (2-x)(x+1) \geq 0$, therefore $u_{n+1} \geq u_n$ after comparison of positive squares. The sequence converges to $\ell \in [1;2]$. After continuity, $\ell = \sqrt{2+\ell}$ implies $(\ell-2)(\ell+1) = 0$; only $\ell = 2$ lies in $[1;2]$.

A4 · Fix a line of reasoning Seq. $a_0 = 1$, $a_{n+1} = 2a_n$ satisfies a relation whose only fixed point is 0. A student concludes $a_n \to 0$. Show the mistake and determine the actual behaviour.

A5 · Independent proof Let $b_0 = 0$, $b_{n+1} = \frac{1}{3}b_n + 2$. Prove $0 \leq b_n < 3$, then that it is increasing. Deduce convergence and determine the limit by the explicit formula or, after S32, by continuity.

2.4 Do not forget negative ratios

First use: S19 and S21.

Property · Behaviour of q^n

If $|q| < 1$, $q^n \to 0$. If $q > 1$, $q^n \to +\infty$. For $q = 1$, the sequence is constant. For $q \leq -1$, it has no limit, not even an infinite one: the signs alternate, and the two subsequences of even and odd indices have incompatible behaviours.

Proof. For $q > 1$, write $q = 1+a$ with $a > 0$. Induction gives $(1+a)^n \geq 1+na$, which tends to $+\infty$; comparison gives the result. For $0 < q < 1$, apply this result to $1/q$ and take the reciprocal. For $-1 < q < 0$, $|q^n| = |q|^n \to 0$ implies $q^n \to 0$ by the squeeze theorem. The case $q = 0$ is immediate from index 1. □

Worked example · Common ratio and behaviour

For $q = -2$, the even-index terms 4^k tend to $+\infty$ and the odd ones -2×4^k to $-\infty$: no common limit. For $q = -1/2$, the signs also alternate, but the absolute value 2^{-n} tends to 0; the sequence converges.

A6 · Compare properties For $(-1)^n$, $-\frac{1}{n+1}$ & n, state: bounded? monotonic? convergent ? Give justifications, not just a table of answers.

One minute, spoken aloud. Explain why "increasing" means neither "positive" nor "tends to $+\infty$ ".

Delayed review · S33 then S66. Go back to identifying the limit by continuity; for a sequence of integrals, rebuild the same chain: bounds, monotonicity, convergence, identification.

Investigation

A limit is not a last term - 30 min.

Consider the sequence (u_n) defined on $\mathbb{N}$ by $u_n = 2 + \frac{3}{n+1}$.

1. Calculate u_0, u_9 and u_{99}. Is there an integer n such that $u_n = 2$? Justify.

2. Determine the smallest integer N such that, for every integer $n \geq N$, $|u_n - 2| < 0{,}01$.

3. Let $\varepsilon > 0$. Give, in terms of ε, an integer N such that $|u_n - 2| < \varepsilon$ for all $n \geq N$.

4. On $\mathbb{N}$ define (v_n) by $v_n = 2 + \frac{(-1)^n}{n+1}$. Prove that it converges to 2 and that it is not monotonic. What does this example show?

Progressive exercises

E1 · Direct application. We define on $\mathbb{N}$ the sequences (u_n) and (v_n) by $u_n = \frac{4n-1}{n+2}$ & $v_n = n^2 - 7n$. Determine their limit, justifying the transformations used.

E2 · Application and justification. We define $u_n = \sqrt{n+1} - \sqrt{n}$ for all $n \in \mathbb{N}$. Using the conjugate expression, determine the limit of (u_n).

E3 · Method to organise. We define $u_n = \frac{\cos n}{n}$ for every integer $n \geq 1$. Bound u_n, then deduce the limit of (u_n).

E4 · Choice of strategy. Seq. (u_n) is increasing and satisfies $u_n < 5$ for all $n \in \mathbb{N}$. Justify that it converges and state what can be said about its limit. Is it necessarily equal to 5? Justify.

E5 · Synthesis or proof. We define $u_n = 3 - 2(-0{,}6)^n$ for all $n \in \mathbb{N}$. Determine the limit of (u_n). Is this sequence monotonic? Justify.

Work from the textbook or a past paper

Barbazo, p. 62, Exercises 17 and 24. In S16: exercise 17, to resolve an indeterminate form by dividing by a power of n. At home after S20: exercise 24, to write up an application of the squeeze theorem.

Homework

DM2 - Approximating a root - 50 min.

Consider $u_0 = 0$ and, for every $n \in \mathbb{N}$, $u_{n+1} = \sqrt{2+u_n}$. Questions 1, 2 and 4 are to be done after S20; question 3 will be revisited after the lesson on continuity in S31.

1. Prove by induction that all terms are defined and that $0 \leq u_n \leq 2$ for every $n \in \mathbb{N}$.

2. Prove by induction that $u_{n+1} \geq u_n$ for all $n \in \mathbb{N}$. Deduce that (u_n) converges.

3. Using the continuity of the function $x \mapsto \sqrt{2+x}$, determine the limit of (u_n).

4. Prove that $0 \leq 2 - u_{n+1} \leq \frac{2-u_n}{2}$ for all $n \in \mathbb{N}$. Deduce an upper bound for $2 - u_n$ in function of n, then an integer N such that $0 \leq 2 - u_n \leq 10^{-3}$ for all $n \geq N$.

MCQ A · definitions and operations

After S16. MCQ B after S20; fixed point after S31.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $u_n = 3 + \frac{2}{n+1}$. Its limit is:

A. 0 B. 2

C. 3 D. $+\infty$

2. $\frac{2n^2+1}{5n^2+n}$ tends to:

A. $\frac{2}{5}$ B. $\frac{5}{2}$

C. 0 D. $+\infty$

3. $u_n \to 2$ and $v_n \to 0$. What can be stated without any further assumption?

A. $u_n / v_n \to +\infty$ B. $u_n v_n \to 0$

C. $u_n / v_n \to 0$ D. $u_n + v_n \to 0$

4. For $u_n = 2 + \frac{3}{n+1}$, the first index from which $|u_n - 2| < 0.1$ is:

A. 28 B. 29

C. 31 D. 30

5. Seq. $(-1)^n$ is:

A. converging to 0 B. bounded and divergent

C. increasing D. diverging to $+\infty$

MCQ B · choosing a theorem

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. We know u_n is increasing and $u_n \leq 5$. Necessarily:

A. $u_n \to 5$ B. $u_n \geq 0$

C. $u_n \to \ell \leq 5$ D. u_n constant

7. $u_n = \frac{\sin n}{n}$ for $n \geq 1$. The direct tool is:

A. monotone convergence

B. a fixed-point equation

C. the law of large numbers

D. the squeeze theorem

8. An increasing sequence that is not bounded above:

A. tends to $+\infty$

B. may tend to 0

C. is always geometric

D. oscillates between two values

9. $\sqrt{n^2+n} - n$ tends to:

A. $+\infty$

B. 1

C. $\frac{1}{2}$

D. 0

10. If $0 \leq u_n \leq (\frac{3}{4})^n$ for every index:

A. u_n is geometric

B. $u_n \to 0$

C. u_n is decreasing

D. $u_n \to \frac{3}{4}$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Every convergent sequence is monotonic.

2. Every increasing bounded sequence converges.

3. If $u_n \to \ell$ & $u_n < 5$, then $\ell < 5$.